

Manuel A. Santana

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EDUCATION

California Institute of Technology,

Sept 2022 - June 2027 (expected)

PhD in Applied and Computational Mathematics

Advisor: Oscar Bruno

Utah State University

August 2016 - May 2022

Bachelor's of Science Magna Cum Laude

Major: Computational Mathematics Minor: Computer Science Minor

Research Articles

Published Articles

Bruno, O. P., **Santana, M. A.** 2026 Efficient time-domain scattering synthesis via frequency-domain singularity subtraction, *Journal of Computational Physics*, p. 115078, 2026.

Bruno, O. P., **Santana, M. A.**, and Trefethen, L.N. 2026. Evaluation of resonances: adaptivity and AAA rational approximation of randomly scalarized boundary integral resolvents. *SIAM Journal on Scientific Computing* 48.3 A1260-A1283.

Brauer A, Krawick M, and **Santana M. A.** 2020. Numerical Analysis of the 1-D Parabolic Optimal Transport Problem, *SIAM SIURO*, Vol 14:150-166

Beasley L, Brown D, Mousley J, and **Santana M. A.** 2023 Cordiality of Digraphs, *Journal of Algebra Combinatorics Discrete Structures and Applications*, Vol 10, Issue 1, pp.1-13

Huynh M, **Santana M. A.** 2021. Alternating Minimization for Computed Tomography with Unknown Geometry Parameters, *SIAM SIURO*, Vol 15:62-76, 2022 DOI:10.1137/21S1441638

Conference Papers:

M. A. Santana, L. N. Trefethen, O. P. Bruno, "Computation of Cavity Resonances via AAA Rational Approximation of Randomly Scalarized Boundary Integral Resolvents", in L. Gizon (Editor), Book of Abstracts, The 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation (WAVES 2024), pp. 19-24 (2024), DOI 10.17617/3.MBE4AA (Edmond MPDL).

Santana, M., Mousley, J., Brown, D., Beasley, L.B. 2024. (2, 3)-Cordial Trees and Paths. In: Hoffman, F., Holliday, S., Rosen, Z., Shahrokhi, F., Wierman, J. (eds) *Combinatorics, Graph Theory and Computing*. SEICCGTC 2021. Springer Proceedings in Mathematics & Statistics, vol 448. Springer, Cham. https://doi.org/10.1007/978-3-031-52969-6_12

Mousley, J.M., Beasley, L.B., **Santana, M.** Brown, D.E. 2024. (2, 3)-Cordial Oriented Hypercubes. In: Hoffman, F., Holliday, S., Rosen, Z., Shahrokhi, F., Wierman, J. (eds) Combinatorics, Graph Theory and Computing. SEICCGTC 2021. Springer Proceedings in Mathematics & Statistics, vol 448. Springer, Cham. https://doi.org/10.1007/978-3-031-52969-6_13

Invited Talks

International Conference on Spectral and High Order Methods (ICOSAHOM) 2025 - Mini-symposia on Novel Methods for wave problems and integral equations

Workshops:

Interactions of geometric spectral theory with numerical methods and applications — Isaac Newton Institute Cambridge England April 13-17 2026

Teaching Experience

Teaching Assistant

August 2023 – Current

California Institute of Technology

Linear Algebra Recitation Leader

August 2020 – August 2021

Awards

Goldwater Scholar

March 2021

NSF Graduate Research Fellowship

April 2022

Undergraduate Researcher of the Year - College of Science, Utah State University

2021

Undergraduate Researcher of the Year - Math Department, Utah State University

2021

Hispanic Scholarship Fund Scholar

May 2020

Utah State University Presidential Scholar

August 2016